Table 12.1 Tissue system and its functions

Tissue System	Components	Functions
Dermal Tissue System	Epidermis and Periderm (in older stems and roots)	• Protection • Prevention of water loss
Ground Tissue System	Parenchyma tissue Collenchyma tissue Sclerenchyma tissue	• Photosynthesis • Food storage • Regeneration • Support • Protection
Vascular Tissue System	Vascular tissues - Xylem tissue - Phloem tissue	• Transport of water and minerals • Transport of food

Cuticle is present on the outer wall of epidermis to check evaporation of water. Trichomes and root hairs are the epidermal outgrowths.

Functions:

i) Epidermis protects the inner tissues.

ii) Stomata helps in transpiration.

iii) Root hairs help in absorption of water and minerals.

12.2.2 Ground Tissue System

It includes all the tissues of the plant body except epidermal and vascular tissues like (i) Cortex (ii) Endodermis (iii) Pericycle (iv) Pith

12.2.3 Vascular Tissue System

It consists of xylem and phloem tissues. They are present in the form of bundles called vascular bundles. Xylem conducts water and minerals to different parts of the plant. Phloem conducts food materials to different parts of the plant.

There are three different types of vascular bundles namely (i) Radial (ii) Conjoint (iii) Concentric

(i) Radial Bundles

Xylem and phloem are present in different radii alternating with each other. e.g. roots

(ii) Conjoint bundles

Xylem and phloem lie on the same radius. There are two types of conjoint bundles.

a) Collateral

Xylem lies towards the centre and phloem lies towards the periphery.

When cambium is present in collateral bundles, it is called open. e.g. dicot stem and collateral bundle without cambium is called closed. e.g. monocot stem.

b) Bicollateral

In this type of bundle, the phloem is present on both outer and inner side of xylem. e.g. *Cucurbita*

(iii) Concentric Bundles

Vascular bundle in which xylem completely surrounds the phloem or viceversa is called concentric vascular bundle. It is of two types:

1. **Amphivasal:** Xylem surrounds phloem. e.g. *Dracaena*
2. **Amphicribral:** Phloem surrounds xylem. e.g. Ferns

Endarch: Protoxylem lies towards the centre and metaxylem lies towards the periphery. e.g. stem.

Exarch : Protoxylem lies towards the periphery and metaxylem lies towards the centre. e.g. roots.

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